Wavelet Based Thermogram Analysis for Breast Cancer Detection

Sourav Pramanik
Department of Computer Science and Engineering
Jadavpur University
Kolkata, India
srv.pramanik03327@gmail.com

Debotosh Bhattacharjee
Department of Computer Science and Engineering
Jadavpur University
Kolkata, India
debotosh@ cse.jdvu.ac.in

Mita Nasipuri
Department of Computer Science and Engineering
Jadavpur University
Kolkata, India
mnasipuri@ cse.jdvu.ac.in

***Abstract*—In this work, we have reported a novel automatic technique to detect early breast cancer by analyzing breast thermogram. The system comprises mainly of three steps: the breast region segmentation from the original image, feature extraction, and finally classification and performance analysis. In the segmentation phase, we have first removed the background region by applying the Otsu's thresholding method followed by a reconstruction technique. Then the inframammary fold is detected to mark the lower limit of the breast. After that, the upper limit of the breast is identified by discerning the axilla. Finally, the breast region is extracted based on these two limits. In the next phase, we have extracted features from the region of interest to identify the early breast cancer, which is the most crucial and significant step. At the very beginning of the feature extraction process, we have computed the initial feature point image (IFI) for each segmented breast thermogram by applying discrete wavelet transform (DWT). After that, different types of features are extracted from the IFI for the diagnosis of breast cancer. Finally, feed-forward artificial neural network with gradient decent training rule is employed here as a classifier. The major problem in this research is the limited collection of publicly available breast thermal databases. At present, there is only one standard breast thermal database available updated by Visual Lab, Fluminense Federal University, Brazil. We are also developing a new database, which will help the researcher to do research in this field. In this work, we have used 306 breast thermograms of 102 patients collected from Visual Lab. In our proposed system, we found the accuracy of 90.48%, whereas sensitivity and specificity were 87.6% and 89.73% respectively.**

Keywords—breast thermogram; initial feature point image; statistical features; feed-forward MLP; PCA

I. INTRODUCTION

The most common type of cancer among the women is breast cancer and is the second leading cause of cancer-related death for the women. So the early detection of a breast tumor, before they metastasize and escalate to the neighboring cells, could minimize mortality rate. There are lots of imaging techniques available in the field of early breast cancer detection. Among them, the most commonly used method, the gold standard, presently called as X-ray mammography. As a result, considerable progress has been made for the development of the automatic diagnostic tool. However, the major problems with X-ray mammography based breast screening are that it is an invasive technique, and the electromagnetic radiation itself helps the growth of breast tumor by 2% during each periodic inspection. Due to the recent advancement in the thermal infrared technology, properly calibrated thermography and apt patient protocols IR systems have received lots of attention for the use of the medical imaging especially in breast cancer detection. Breast thermography can be a prospective early breast cancer detection method if it uses appropriate patient protocols [1]. The infrared thermal imaging records the temperature distribution of the body surface to form an image (or thermogram). In [19], the Greek physician, Hippocrates wrote that "*In whatever part of the body excess of heat or cold is felt, the disease is there to be discovered*". Cancerous cells usually produce more heat than the normal cells because of their higher metabolic activity, angiogenesis and vascular dilation [21]. Thus, the heat and blood vessel activity is the indication of the existence of cancer cells in their premature stage of growths [2]. The pathological procedures for increased temperature in breast cancer cells are that the breast cancer cells produce nitric oxide (NO), which restrains with the normal nervous system of the breast blood vessel flow and as a result a local vasodilatation in the early phases of cancerous cell growth escalating the angiogenesis in later phases. The subsequently increased blood flow, in that area, causes a temperature increase in comparison to the normal breast tissue temperature. Even deep breast lesions seem to have potential to influence the changes in the skin temperature. Since cancerous cells generate more heat than normal cells, the thermal breast patterns, which contain important information about the goodness of health and type of the tumors [10, 13-14]. The results of thermography could be able to apprise women of breast cancer up to 8-10 years before mammography can detect [2]. Apart from that, it has also

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some other benefits, such as it is cheap, non-invasive, painless, non-radiating, fast, and passive.

Any system for early breast cancer detection can be divided into mainly three steps: a region of interest segmentation, feature extraction, classification and performance analysis. From past few years, some significant amount of works has been done on early breast cancer detection. Since it is very difficult to extract breast boundary in the thermal images due to the lack of clear structure or focus, some authors prefer manual segmentation. In [3], authors manually segmented the region of interest with the help of a doctor. However, for the diagnosis of breast cancer, they have used statistical features and Fuzzy Classifier. Qi et al. [4] segmented region of interest based on edge detection followed by Hough transformation. In their method, three points have been computed for the extraction of the breast region. Those three points, namely two armpit points (derived from the largest curvature point of the left and right body boundaries) and the central point of two breasts (calculated from the intersection point between two parabolas of the lower breast boundary). After that they have measured the asymmetry between two breasts using two different techniques: asymmetry measure using unsupervised learning (used k-means algorithm), and asymmetry measure using supervised learning (used K-NN classifier). Tang et al. [5] neither segmented region of interest by manually nor automatic but used an original image for diagnosis of the breast cancer. They have measured the localized surface temperature increased on the breast surface for detection of breast cancer. Segmentation using edge detection followed by parabolic Hough transformation reported in [6]. Authors first reduced the noise using 6 × 6 Gaussian filter then Hough transformation is applied to decrease the computational cost. Finally, for the diagnosis of the breast cancer they have extracted four features (average, variance, skewness, and kurtosis). In [7], authors did not report any segmentation results, they have measured only two features (such as skewness and kurtosis) for the diagnosis of breast cancer. In [8], breast region is segmented by first extracting the body boundary using Canny edge detection and elimination of inner edges. Then Fuzzy c-means clustering algorithm is used for segmentation of the breast region. After that, Radon projection is applied to these segmented images for the extraction of bispectral invariant features. Finally, Adaboost classifier is used to classify images with malignant, benign and normal classes. They have shown that their approach detected malignant cases with 95% accuracy. In [9], authors proposed a complementary learning Fuzzy neural network, called FALCON-ART for the analysis of the breast thermogram. However, they did not mention any segmentation technique or results. In [10], authors provide details review on breast thermogram segmentation, feature extraction, classification and performance analysis.

The research on the breast cancer is still in its early stage, and there are no such robust segmentation methods, and optimum feature extraction techniques are available. Hence, in this paper, we have proposed automatic segmentation technique and a discrete wavelet transform driven multiple features extraction methods for the diagnosis of breast cancer. For the segmentation of the breast region, we have first identified lower limit and an upper limit of the breast. Based on these two limits, breast region is segmented. After that, thermal breast patterns are analyzed to identify the cancerous patients. Here, these patterns are analyzed in multi-resolution scale by applying discrete wavelet transform (db1) and then initial feature point image (IFI) is calculated. After that, multiple features are extracted from the IFI for the diagnosis of the breast cancer. To validate the proposed method, we have tested on a database, which consists of a total 306 thermograms of 102 individuals. After feature extraction, feed-forward neural network with gradient decent training rule and two hidden layers (30 and 15 hidden nodes) is used for the classification of healthy and unhealthy patients. Based on the obtained classification results, we have computed three performance measures: accuracy, sensitivity, and specificity. Our system offers accuracy of 90.48%, the sensitivity of 87.6% and specificity of 89.73% that is better than the methods proposed in [15] and [20].

The rest of the paper is organized as follows: Section 2 describes proposed algorithm in detail, followed by the results obtained along with performance analysis in Section 3 and finally Section 4 concludes the paper.

II. Proposed Method

In this section, we have presented a detailed algorithm for breast cancer detection using thermal breast images. Since it is very difficult to extract breast boundary in the thermal images due to the lack of clear structure or focus, we have first tried to extract the breast region from the original captured thermal images. Then we have analyzed the heat patterns in the breast region to detect breast cancer. The proposed method comprises of mainly four steps: pre-processing, segmentation, feature extraction and classification.

A. Pre-processing

The FLIR SC-620 thermal infrared camera has been used to capture the breast thermograms. All the captured thermal breast images were digitized into 600×800 pixels with a resolution of 24-bit per pixel. Original breast thermograms also consist of some labels and tags, which are embedded in the vendor software, as shown in fig.1a. Thus, the very first step of the proposed method is to remove all those labels and tags and then converted into 8-bit grayscale counterparts. Dedicated vendor software has been used to remove all those labels and tags except heat bar. Finally, 24-bit color images are converted into a corresponding 8-bit grayscale image using National Television System Committee (NTSC) conversion formula (equation 1).

$$I_{gray} = \left(0.299 * red_{component}\right) + \left(0.587 * green_{component}\right) + \left(0.114 * blue_{component}\right) \quad (1)$$

where I_{gray} is the 8-bit gray scale image.

B. Segmentation

As discussed earlier, the breast thermograms are acquired in a large area, which includes upper body area, breast area, and lower body part. Therefore, it is necessary to segment the breast region part from the rest of the body. For the segmentation of the breast region, we have first identified lower limit and an upper limit of the breast region. Then, based on these two limits, breast region is segmented. Our region of interest segmentation process is divided into three steps: background removal, inframammary fold (lower limit) detection, and axilla detection (upper limit).

1) Background removal

Original breast thermal images consist of the background region, which can mislead the segmentation method or feature extraction process. In this section, we have presented a method for the removal of the background from the breast thermogram. For the removal of the background, we have employed here Otsu's thresholding method followed by a gray level reconstruction technique. The details Otsu's algorithm is described in [11]. When we apply Otsu's thresholding method on the thermal breast images, it removes background accurately but some undesired effects occur in the object (foreground) region, as shown in fig.2b. These undesired effects can mislead the next step of the segmentation process. Thus, to overcome this problem, all the white pixels were replaced by the original image pixels. It may be noted that during thresholding background pixels were replaced by 0 (black) and foreground pixels were replaced by 1 (white). As discussed earlier, the undesired effects on the object region, here black pixels, is then removed by applying the proposed gray level reconstruction technique on the initial background removed image. The details reconstruction technique is as follows:

Algorithm 1: Removal of undesired effects from $\boldsymbol{I_o}$

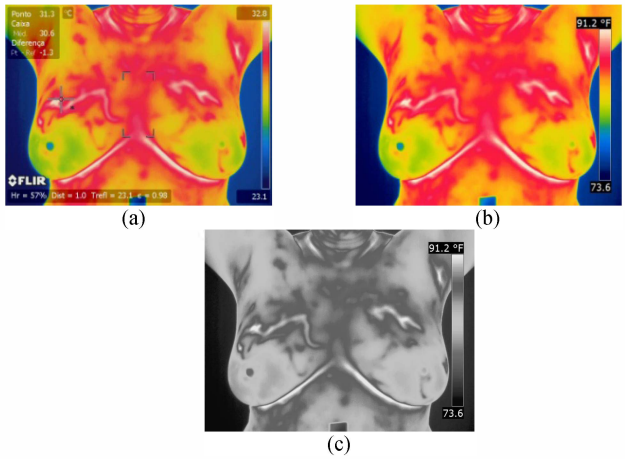
Let I be the original grayscale image and $\boldsymbol{I_o}$ be the initial background removed image after applying Otsu's threshold.
Input: $\boldsymbol{I_o}$
Output: Complete background removed image $\boldsymbol{I_c}$

Step 1: Vertically partition $\boldsymbol{I_o}$ into two equal parts, say *Left_half* and *Right_half*.

Step 2: Consider *Left_half* and scanned horizontally from left-to-right. Repeat the process until all the black pixels in the object region are replaced by original image pixel values.

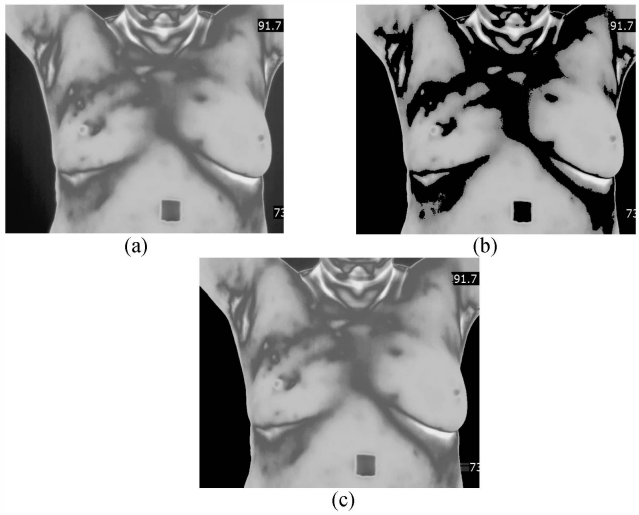
- Each pixel of *Left_half* is replaced by original image pixel value if its value is zero and immediate predecessor value is non-zero.

Step 3: Consider *Right_half* and scanned horizontally from right-to-left. Continue the same process as mentioned in Step-2.
Step 4: Horizontally concatenate the processed two halves obtained in Step-2 and Step-3 to obtain final background removed image $\boldsymbol{I_c}$.



(a) (b) (c)

Fig. 1 (a) The original image, (b) after processing using FLIR software, (c) gray image



(a) (b) (c)

Fig. 2 (a) Original gray image, (b) after applying Otsu's threshold, (c) after applying our reconstruction technique on (b)

2) Lower limit detection

After removal of background correctly, our next task is to find out the lower limit of the breast in order to remove the lower body part. To find out that limit, the hypothesis that we have followed here is that *the maximum temperature occurs in the inframammary fold region of the lower part of the breast.* As we know that the maximum temperature causes the highest pixel value in the grayscale thermal image. Thus, we have first horizontally partitioned the background removed image I_c into two equal sized parts, which we call as *top_half* and *bottom_half*. The *bottom_half* consists of the lower breast region and considered as our current working image. The very first step of the lower limit detection is to convert the *bottom_half* image into binary. Thus, first a maximum value

has been calculated, say *max_val*, from *bottom_half* image. Then the pixels with the value greater or equal to the *max_val* are replaced by 1, whereas pixels replaced by 0 for those have the value less than the *max_val*. In the obtained binary image, inframammary region reflects as a thick parabolic curve, as shown in fig. 3e. But from this thick parabolic curve, it is quite difficult to find a pixel which has maximum *x-coordinate* value. Hence, we have used here morphological thinning [17] operation to procure one pixel thin parabolic curve, as shown in fig. 3f. Thermal breast images may consist some sort of noise which can mislead the process of finding the pixel having maximum *x-coordinate* value. Thus, before applying morphological thinning operation, we have first removed noise using morphological dilation operation followed by a median filter. Now our task is to find out the white pixel which has maximum *x-coordinate* value. That can be obtained if we search vertically top-to-bottom from left-to-right or right-to-left and consider the first white pixel coordinate. But due to the lack of clear structure or focus of the lower breast region, sometimes our thresholding method is unable to produce two breast parabolas at the same time. Thus, if an image consists of only right parabola and if we search from left-to-right then searching will take an enormous time and also will give wrong lower limit. Hence, in order to avoid this problem, first vertically divide the image into four equal parts (LH1, LH2, RH1, and RH2), as shown in fig. 3i. Then the pixel having maximum *x-coordinate* value is calculated. The detailed algorithm for lower limit identification as follows:

Algorithm 2: Breast lower limit detection

Input: bottom_half
Output: Pixel that is having maximum *x-coordinate* value.

Step 1: Find out the maximum pixel value and call as *max_val*.
Step 2: Based on the computed *max_val*, convert the image from gray to binary.
Step 3: Apply morphological dilation operation.
Step 4: Perform the filtering operation by applying the median filter.
Step 5: Perform the morphological thinning operation to obtain thin parabolic curve.
Step 6: Search for a white pixel that has maximum *x-coordinate* value.

(i) Vertically divide the image obtained in step-5 into four equal parts, namely LH1, LH2, RH1, and RH2.
(ii) Consider LH1 and scanned vertically from *right-to-left* and find the white pixel with maximum x-coordinate value, say *LH1_val*. If found then go to step *(vi)*, otherwise go to step *(iii)*.
(iii) Vertically scan LH2 from *left-to-right* and find the white pixel that has maximum *x-coordinate* value, say *LH2_val*. If there is no such pixel is found then go to step *(iv)*, otherwise go to step *(vi)*.
(iv) Vertically scan RH1 from *left-to-right* to find out a white pixel that has maximum *x-coordinate* value, say *RH1_val*. If found then go to step *(vi)*, otherwise go to step *(v)*.
(v) Vertically scan RH2 from *right-to-left* and find the white pixel that has maximum *x-coordinate* value, say *RH2_val*. If found then go to step *(vi)*.
(vi) Comparison :
IF RH1_val ≠NULL, then
If LH1_val=LH2_val=NULL, then max_x-coordinate=RH1_val
Else if LH1_val>=RH1_val, then max_x-coordinate=LH1_val
Else if LH2_val>=RH1_val, then max_x-coordinate=LH2_val
Else
If LH1_val=LH2_val=NULL, then max_x-coordinate=RH2_val
Else if LH1_val>=RH2_val, then max_x-coordinate=LH1_val
Else if LH2_val>=RH2_val, then max_x-coordinate=LH2_val

3) Upper limit detection

The upper limit of the breast is obtained from the *top_half* breast image by calculating the maximum curvature point. Maximum curvature point is calculated from the body boundary of the *top_half* breast image. Body boundary is extracted by applying Canny edge detector followed by some post-processing operation. Because due to the different metabolic activity in the human body which reflects on the breast thermal images, Canny edge operator detects enormous edges. Thus, it is difficult to extract body boundary from that edge image. Hence, we have done some pre-processing and post-processing operation before and after applying the Canny edge detector to extract the body boundary. Fig.3h shows the body boundary extracted by our algorithm. The detailed algorithm for identification of the upper limit of the breast as follows:

Algorithm 3: Breast upper limit detection

Input: top_half
Output: Maximum curvature point

Step 1: Compute the global threshold value of the *top_half* thermogram using Otsu's method.
Step 2: Based on the computed threshold value, convert it to a binary image.
Step 3: Take the complement of obtained binary image.
Step 4: Fill all the holes using morphological filling operation.

Step 5: Take the complement of the image obtained in step-4.
Step 6: Apply the Canny edge detector on the resultant image.
Step 7: Keep only the perimeter pixel (used 8-connectivity).
Step 8: Extract the body boundaries using the following steps:

1. Vertically divide the image that is obtained in step-7 into two equal parts (say *left_im* and *right_im*).
2. Consider *left_im* and scanned horizontally from left-to-right. In each iteration, search for first white pixel, if found then remaining white pixels are replaced with black (0) pixels.
3. Next consider the *right_im* and scanned horizontally from right-to-left and continue the same process as mentioned in step (2).
4. Finally, take the horizontal concatenation of processed two halves to get final body boundary image.

Step 9: Compute the maximum curvature point and consider as an upper limit (axillary point).

After two limits detection, breast regions are segmented, as shown in Fig. 3j. However, due to the different sizes of the breasts for a different person, the size of the breast region will also be different. Thus to keep all the breast region size equal, we have plus or minus the upper limit row. In our implementation, each original image size is 600 × 800. After segmentation, each image size becomes 384 × 672.

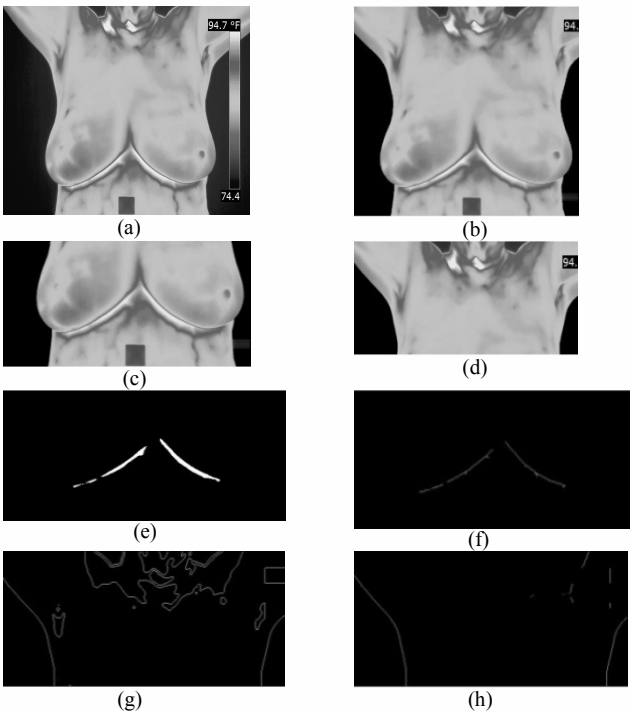


Fig. 3 (a) original grayscale image, (b) corresponding background removed image, (c) bottom half, (d) top half, (e) after thresholding of bottom half image, (f) corresponding thinning image of (e), (g) edge image of (d), (h) boundary extracted image of (g), (i) thin image is divided into four equal parts, (j) final segmented image.

C. Feature Extraction

The feature extraction is the most critical phase of the diagnosis of breast cancer. Usually, the thermal breast image exhibits the complex structure of tumor in the form of temperature distribution that is quite different from the temperature distribution of the background breast tissue. The temperature increased in the tumor area influence the changes in the skin temperature and made a thermal pattern that is the necessary information about the goodness of health. By observing the breast thermogram used in this work, we note that the gray level of the breast thermogram changes abruptly from low to high or high to low temperature, as shown in fig. 4(a). Although, it is hard to analyze these gray levels changes in a single resolution level. Thus, we have first applied the discrete wavelet transform (db1) to obtain the details structural information of the tumor region. The wavelet transform is a well-known multiresolution analysis tool, which decomposes source images into different resolutions. At different resolution levels, the information of an image usually represents different physical structures present in the image. The detail coefficients of wavelet transform usually reflect the structural image information in details that are similar to looking a nearby object. On the other hand, approximation coefficient exhibit a coarser resolution that is similar to looking a far object that possesses gross information. Thus, we considered only the detail coefficients of the wavelet transform of breast thermogram and computed the Initial Feature point Image (IFI), as shown in fig. 4(b), for the diagnosis of breast cancer. Let us have three detail coefficients LH (horizontal), HL (vertical), and HH (diagonal). The IFI is computed using the equation (2).

$$IFI = \sqrt{LH^2 + HL^2 + HH^2} \qquad (2)$$

Since the number of feature points in IFI is larger, the direct use of IFI will cause the higher verification rate. Thus, it is necessary to select the important features to represent the IFI. Thus, the mean value of $\boldsymbol{x}$ and $\boldsymbol{y}$ coordinates of feature points, some statistical features of IFI, and feature points count are considered as the features. The feature vector V for each breast thermogram is given as follows:

$$V = [M_x, M_y, SF_{IFI}, C] \qquad (3)$$

where M_x and M_y are the mean values of feature point coordinates $\boldsymbol{x}$ and $\boldsymbol{y}$, SF_{IFI} is the statistical features of IFI and C is the number of feature points in IFI.

1) Calculation of M_x and M_y

The mean value of x and y coordinates of feature points are calculated as follows:

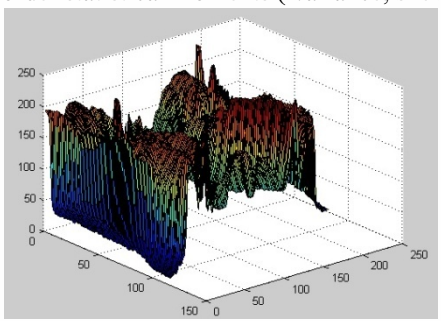
$$M_x = \frac{1}{n}\sum_{i\in R}^{MN} x_i \qquad (4)$$

where M_x is the mean value of x-coordinate of the feature point, n is number of non-zero intensity, R is the region having non-zero intensity, M and N are the dimensions of the IFI. Similarly, M_y is computed using equation (5).

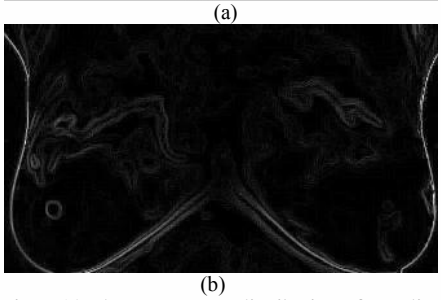
$$M_y = \frac{1}{n}\sum_{i\in R}^{MN} y_i \qquad (5)$$

2) Statistical feature extraction

Since the local variations of the image intensity are very much important, the statistical features are extracted locally from the IFI. Thus, IFI is first divided into n number of equal sized (16 ×16) non-overlapping blocks $B_1, B_2, \ldots, B_n$ *(where n=1,2,....,252)*. In this work, each segmented breast thermogram size is 384×672 and after wavelet transform it becomes 192×336. Thereafter, features $f_1, f_2, \ldots, f_m$ *(where m=1,2,...,5)* are extracted from each block. In this work, we have computed the mean, variance, entropy, skewness, and kurtosis and considered those as features. The mean, variance, skewness, kurtosis, and entropy are computed using the equations (6), (7), and (8) [12]. As we know that the higher order statistical moments (variance, skewness, and kurtosis) describes very precisely the characteristic of a distribution, assuming that each patch represents a random distribution. Finally, all the block features are organized in a row vector and stored in a matrix K. In our implementation, we have considered three different views of each thermogram, which are a frontal view, 90 degrees on the right side, and 90 degrees on the left side. Fig.5 shows the examples of frontal view, 90 degrees on the right side and 90 degrees on the left side of patient images. Thus, for each patient features are organized in a matrix $K_{3\times1260}$. For any classifier, the high dimension of the feature vectors may create difficulty during the learning process and as a result it may cause high variance. Therfore, high input features dimensionality usually demands tuning of the classifier so that the variance should be low and bias should be high. There are lots of feature selection tecniques available, which can discard the irrelevent features and keep the relevent one. Among them principal component analysis (PCA) [16] is the most popular one due to its simplicity and accuracy. Hence, in this work we have used PCA for the dimentionality reduction of each patient feature matrix $K^r_{3\times1260}$, where r is the number of patients. For each patient, we consider two eigen vectors corresponding to two highest eigen values. Thus, total 3780 features for a patient is now represented with only six features.

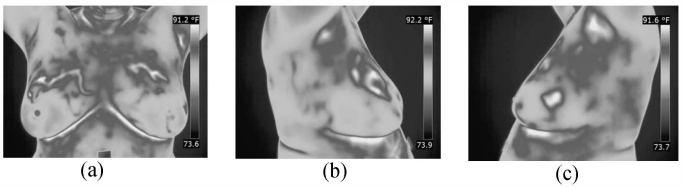


(a)



(b)

Fig. 4 (a) The temperature distribution of a malignant breast thermogram, (b) Initial feature point image.



(a) (b) (c)

Fig. 5 (a) Frontal view, (b) 90 degree of the right side, (c) 90 degree on the left side.

$$m^{(1)} = \frac{1}{N}\sum_{i=1}^{N} x_i \qquad (6)$$

$$m^{(q)} = \frac{1}{N}\sum_{i=1}^{N}\left(x_i - m^{(1)}\right)^q \qquad (7)$$

$$Entropy = -\sum_{i=1}^{N} x_i \log x_i \qquad (8)$$

where, N is the number of pixels in the specified block and q=2,..,4.

D. Classification

An artificial neural network is one of the most popular supervised classifier, which consists of two operations: training and testing. The most widely used neural classifier is feed-forward multilayer perceptron network (MLP). The general structure of this network consists of an input layer, hidden layer, and an output layer. In this work, a four layer feed-forward perceptron has been developed which has 15 neurons in the input layer, 30 neurons in the first hidden layer, 15 neurons in the second hidden layer and one neuron in the output layer. Total 156 samples of 52 individuals are used to train the network and remaining 150 samples of 50 individuals are used to test and validate the system. Since the performance of the classifier entirely depends on the initial training, we have taken a good mixture of healthy and unhealthy breast thermal features and used gradient decent training rule.

III. Experimental Results and Discussion

Our proposed method is tested on a database, which consists of a total 306 thermograms of 102 individuals (123 for unhealthy and 183 for healthy patients), those are collected

from Visual Lab, Fluminense Federal University, Brazil [18]. For each patient, we have considered three different views thermograms, which are frontal view, 90 degrees on the right side, and 90 degrees on the left side and the total of 15 features are extracted which have been discussed in the previous section. Before extraction of features, the breast regions are segmented from the original image using our proposed segmentation method. Fig. 2(j) shows the segmented image corresponding to the original image. In our implementation, each original image size is 600 × 800. After segmentation, each image size becomes 384 × 672. Since the breast thermograms are acquired in a large area, it is necessary to segment the breast region part from the rest of the body. Experimental result shows that our method has segmented breast region significantly well. During the features extraction process, initial feature points image (IFI) is divided into small non-overlapping blocks (16 × 16) and extracted five statistical features from each block. Here, we have considered the small block size because to analyze the small tumor region separately. In the breast, since normal breast tissue patterns are available, the large block will overlap the small tumor pattern with normal breast tissue pattern. Some higher order statistical moments (variance, skewness and kurtosis) have been computed here as features because as we know that the higher order statistical moments describe the characteristic of a distribution very precisely (assuming that each block represents a random distribution). After feature extraction, we then train feedforward neural network with 156 samples of 52 individuals (63 unhealthy and 93 healthy samples) and remaining 150 samples of 50 individuals (60 unhealthy and 90 healthy samples) are used for validation. Based on the obtained classification results, we have computed three performance measures: accuracy, sensitivity and specificity. Accuracy, sensitivity and specificity are calculated using the equation (9), (10) and (11) [10]. The obtained accuracy is 90.48%, sensitivity is 87.6% and specificity is 89.73% which are better than the approaches mentioned in [15] and [20]. They have reported the sensitivity is about 81.2% and specificity 88.2% but not measured the classification accuracy. In [20], authors have been reported the accuracy of 88.10%, whereas sensitivity and specificity are of 85.71% and 90.48% respectively.

$$Accuracy = \frac{(TP + TN)}{(TP + FP + FN + TN)} \quad (9)$$

$$Sensitivity = \frac{TP}{TP + FN} \quad (10)$$

$$Specificity = \frac{TN}{TN + FP} \quad (11)$$

where TP=true positive, FN= false negative, TN=true negative, FP= false positive

IV. Conclusion

In this paper, we have presented an automatic computer-aided diagnosis approach for the detection of breast cancer using thermal breast imaging that can be used as a second opinion. In this work our contribution includes: 1) an automatic segmentation method, which has been devised for the extraction of breast region from the captured original breast thermogram, 2) a new feature extraction technique based on multiresolution analysis is employed. Finally, features are fed into the feedforward neural network classifier for the diagnostic prediction (healthy or unhealthy) of the examined patient. The experimental results show that the system achieved accuracy of 90.48%, the sensitivity of 87.6% and specificity of 89.73% that are comparable to that obtained by some other existing method. Apart from that our proposed method is very simple and efficient one.

For the segmentation, two limits (upper and lower) are identified automatically and based on those breast thermograms are segmented. Thus, it consists of some non-breast region that could not be removed in this step. After segmentation, features are extracted from the entire breast region. Since thermal tumor patterns in the breast region play a significant role in the diagnosis of early breast cancer, it is important to identify that tumor region only. In future, we will try to address the entire problem mentioned here along with some suitable soft computing technique to achieve better accuracy.

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